ANALYSIS REPORT

Track 1 Remaining Family Cellwise Final Closeout Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the full remaining-family exact-paper `Track 1` cellwise batch prepared in:

- `doc/reports/campaign_plans/track1/exact_paper/2026-04-18-22-28-04_track1_remaining_family_cellwise_reference_campaigns_plan_report.md`

  The batch replicated the accepted `SVM` workflow family-by-family by training a separate exact-paper model for every paper-facing cell of the remaining `Track 1` families.
- completed family campaigns: `9`
- completed validation runs: `171`
- failed validation runs: `0`
- aggregate campaign artifact root:

  `output/training_campaigns/track1_remaining_family_cellwise_reference_campaigns_2026_04_18_22_28_04/`

Operational Outcome

- every non- `SVM` exact-paper family now has its full `19` -run cellwise bank
- all `9` family campaign folders now expose:
  - `campaign_leaderboard.yaml`
  - `campaign_best_run.yaml`
  - `campaign_best_run.md`
- the aggregate remaining-family campaign root now exposes the final `171` -entry

  leaderboard and deterministic bookkeeping representative
- `doc/running/active_training_campaign.yaml` now records the canonical results

  report path for the completed batch

Family Recovery Outcome

Family	Amplitude Run	Phase Run	Best Run	Best Scope	Best Closure Score	Open Cells
MLP	track1_mlp_amplitude_162_cellwise_reference	track1_mlp_phase_156_cellwise_reference	track1_mlp_amplitude_162_cellwise_reference	amplitudes_only	1.000	11
RF	track1_rf_amplitude_162_cellwise_reference	track1_rf_phase_156_cellwise_reference	track1_rf_amplitude_162_cellwise_reference	amplitudes_only	1.000	1
DT	track1_dt_amplitude_240_cellwise_reference	track1_dt_phase_156_cellwise_reference	track1_dt_amplitude_240_cellwise_reference	amplitudes_only	1.000	4
ET	track1_et_amplitude_0_cellwise_reference	track1_et_phase_156_cellwise_reference	track1_et_amplitude_0_cellwise_reference	amplitudes_only	1.000	5
ERT	track1_ert_amplitude_1_cellwise_reference	track1_ert_phase_156_cellwise_reference	track1_ert_amplitude_1_cellwise_reference	amplitudes_only	1.000	6
GBM	track1_gbm_amplitude_162_cellwise_reference	track1_gbm_phase_156_cellwise_reference	track1_gbm_amplitude_162_cellwise_reference	amplitudes_only	1.000	1
HGBM	track1_hgbm_amplitude_162_cellwise_reference	track1_hgbm_phase_156_cellwise_reference	track1_hgbm_amplitude_162_cellwise_reference	amplitudes_only	1.000	1
XGBM	track1_xgbm_amplitude_162_cellwise_reference	track1_xgbm_phase_156_cellwise_reference	track1_xgbm_amplitude_162_cellwise_reference	amplitudes_only	1.000	3
LGBM	track1_lgbm_amplitude_162_cellwise_reference	track1_lgbm_phase_156_cellwise_reference	track1_lgbm_amplitude_162_cellwise_reference	amplitudes_only	1.000	1

Aggregate Ranking

Rank	Run	Family	Scope	Paper Cell	Met	Near	Open	Closure Score
1	track1_dt_amplitude_240_cellwise_reference	DT	amplitudes_only	2	2	0	0	1.000
2	track1_dt_amplitude_39_cellwise_reference	DT	amplitudes_only	2	2	0	0	1.000
3	track1_dt_amplitude_40_cellwise_reference	DT	amplitudes_only	2	2	0	0	1.000
4	track1_dt_amplitude_78_cellwise_reference	DT	amplitudes_only	2	2	0	0	1.000
5	track1_dt_phase_156_cellwise_reference	DT	phases_only	2	2	0	0	1.000
6	track1_dt_phase_39_cellwise_reference	DT	phases_only	2	2	0	0	1.000
7	track1_dt_phase_3_cellwise_reference	DT	phases_only	2	2	0	0	1.000
8	track1_dt_phase_78_cellwise_reference	DT	phases_only	2	2	0	0	1.000
9	track1_ert_amplitude_1_cellwise_reference	ERT	amplitudes_only	2	2	0	0	1.000
10	track1_ert_amplitude_39_cellwise_reference	ERT	amplitudes_only	2	2	0	0	1.000
11	track1_ert_amplitude_3_cellwise_reference	ERT	amplitudes_only	2	2	0	0	1.000
12	track1_ert_amplitude_40_cellwise_reference	ERT	amplitudes_only	2	2	0	0	1.000
13	track1_ert_amplitude_78_cellwise_reference	ERT	amplitudes_only	2	2	0	0	1.000

Rank	Run	Family	Scope	Paper Cell	Met	Near	Open	Closure Score
14	track1_ert_phase_156_cellwise_reference	ERT	phases_only	2	2	0	0	1.000
15	track1_ert_phase_40_cellwise_reference	ERT	phases_only	2	2	0	0	1.000

Final Batch Winner

The aggregate batch bookkeeping representative is `track1_dt_amplitude_240_cellwise_reference` from family `DT`.

This is a deterministic winner under the per-run closure-first policy across the full `171`-entry batch. It should be read as a bookkeeping representative of the completed campaign, not as the whole scientific verdict of the promoted cellwise family rows.

- winning scope: `amplitudes_only`
- closure score: `1.000`
- met paper cells: `2`
- near paper cells: `0`
- open paper cells: `0`

Scientific Outcome

- the `171`-run batch successfully materializes the intended `19`-model cellwise banks for `MLP`, `RF`, `DT`, `ET`, `ERT`, `GBM`, `HGBM`, `XGBM`, and `LGBM`
- the refreshed family rows improve the canonical amplitude `RMSE` envelope from `6/10` to `7/10`
- the refreshed family rows leave amplitude `MAE` at `3/10` and phase `MAE` at `5/9`
- the refreshed family rows reduce phase `RMSE` closure to `4/9` because the promoted cellwise references prioritize per-family row fidelity rather than preserving the older full-matrix winner envelope
- under the canonical Table `6` family-alignment rule, no harmonic is yet fully closed after this refresh

Final Benchmark Refresh Effect

The canonical benchmark after the cellwise remaining-family closeout is:

- Table `2` amplitude `MAE`: `3/10` harmonics meet or beat the paper target
- Table `3` amplitude `RMSE`: `7/10` harmonics meet or beat the paper target
- Table `4` phase `MAE`: `5/9` harmonics meet or beat the paper target
- Table `5` phase `RMSE`: `4/9` harmonics meet or beat the paper target
- harmonic-level Table `6` closure: `0/10` fully matched, `8/10` partially matched, `2/10` not yet matched

  The highest-priority still-open harmonics are now:
- `3`
- `240`
- `0`
- `1`
- `156`
- `162`

Final Interpretation

This batch is operationally complete and materially stronger as a reference-bank package than the earlier full-matrix remaining-family pass.

The key final conclusion is

- the remaining-family cellwise batch is fully closed out;
- the repository now has the intended `171` exact-paper reference runs for the nine non-`SVM` families;
- the canonical family rows and the benchmark tables now read from those promoted cellwise references;
- `Track 1` still remains canonically open because the best numeric families do not yet align cleanly with the paper family selections in Table `6`.